

Model Strength x Evolution Factorial Runbook

This runbook implements the final crossed experiment requested by the supervisor:

3 task families x 3 model tiers x 5 evolution techniques, with fixed train/held-out splits and budget-matched controls.

Use the branch model-strength-factorial. Activate your environment first:

```
conda activate Python3_14
```

Smoke Test

The smoke test is no-cost because it uses the builtin game policy generator. It writes to DO NOT COMMIT/, not to official artifacts.

```
$stamp = Get-Date -Format "yyyyMMdd_HH:mm:ss"
python run_model_strength_factorial.py --config configs/model_strength_factorial_smoke.yaml --timestamp $stamp --task-family simple_games --model-tier weak_model --technique single_shot,budget_matched_no_replay
python aggregate_model_strength_factorial.py --run-root "DO NOT COMMIT/model_strength_factorial_smoke/$stamp"
```

Confirm these files exist:

```
Test-Path "DO NOT COMMIT/model_strength_factorial_smoke/$stamp/all_runs_long.csv"
Test-Path "DO NOT COMMIT/model_strength_factorial_smoke/$stamp/cell_means.csv"
Test-Path "DO NOT COMMIT/model_strength_factorial_smoke/$stamp/variance_decomposition/variance_partition_summary.csv"
Test-Path "DO NOT COMMIT/model_strength_factorial_smoke/$stamp/final_report.md"
```

Full Paid Campaign

The full campaign writes official artifacts under runs/cross_family_model_x_evolution_factorial/YYYYMMDD_HHMMSS/.

```
$stamp = Get-Date -Format "yyyyMMdd_HH:mm:ss"
python run_model_strength_factorial.py --config configs/model_strength_factorial.yaml --timestamp $stamp
python aggregate_model_strength_factorial.py --run-root "runs/cross_family_model_x_evolution_factorial/$stamp"
```

If the campaign is interrupted, rerun the same command with the same \$stamp and --skip-existing:

```
python run_model_strength_factorial.py --config configs/model_strength_factorial.yaml --timestamp $stamp --skip-existing
python aggregate_model_strength_factorial.py --run-root "runs/cross_family_model_x_evolution_factorial/$stamp"
```

Planned Scale

The committed model tiers are:

- weak_model: gpt-4.1-nano
- medium_model: gpt-5-nano
- strong_model: gpt-5.4-mini

The planned full design produces 750 run rows:

- simple_games: 3 models x 5 techniques x 10 seeds = 150 rows.
- tsp: 3 models x 5 techniques x 20 seeds = 300 rows.
- cvrp_phase9_real_world: 3 models x 5 techniques x 20 seeds = 300 rows.

Candidate-generation budget is 15,990 total candidates:

- simple_games: 3 models x 10 seeds x (1 + 4 x 100) = 12,030 candidates.
- tsp: 3 models x 20 seeds x (1 + 4 x 8) = 1,980 candidates.
- cvrp_phase9_real_world: 3 models x 20 seeds x (1 + 4 x 8) = 1,980 candidates.

This run is expensive because it separates base model tier from search budget and replay technique. Interpret replay as adding independent value only if it improves over budget_matched_no_replay, not merely over single_shot.

The committed strong tier is a budget-feasible ladder point, not the absolute strongest accessible model. This keeps the full crossed campaign affordable while preserving a transparent weak/medium/strong comparison.

Required Outputs

The runner and aggregator together write:

- all_runs_long.csv
- cell_means.csv
- model_strength_table.csv
- model_x_evolution_matrices/*.csv
- model_x_evolution_matrices/*.png
- variance_decomposition/task_specific_anova.csv
- variance_decomposition/pooled_anova.csv
- variance_decomposition/variance_partition_summary.csv
- effect_sizes/evolution_vs_single_shot_effects.csv
- effect_sizes/evolution_vs_budget_matched_effects.csv
- figures/*.png
- final_report.md
- final_report.pdf

The final report prints the paths to final_report.md, all_runs_long.csv, and variance_partition_summary.csv, plus the empirical conclusion paragraph.

Model Score Policy

The committed model scores are ordinal proxies: weak=1, medium=2, strong=3. This is deliberate because comparable public coding-benchmark scores are not reliably available for every project-specific model alias. If you later obtain comparable HumanEval, MBPP, LiveCodeBench, or SWE-bench Verified scores for all three tiers, update configs/model_strength_factorial.yaml before launching the official campaign and keep benchmark_score_source transparent.